The Immersive Virtual Environment Design Studio

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Abstract. This chapter discusses the implications of designing, perception, comprehension, communication and collaboration within the framework of an 'Immersive Virtual Environment Design Studio'. It has been suggested that the unique properties of immersive virtual environments can empower designers to express, explore and convey their imagination more easily. For these reasons the very different nature of virtual environments may allow architects to create novel designs that make use of additional properties that other realms cannot offer. An architectural design studio was conducted to investigate the relative effectiveness of immersive virtual environment as environment for creation, interpretation, communication of spatial design, and collaboration within a design team. The outcomes of the design studio present how immersive virtual environments successfully aid architectural designing.

Keywords: architectural design, collaboration, design medium, immersive virtual environments.

1 Introduction

There is a distance between the imagination of a design and its representation, communication and realisation: architects use a variety of tools to bridge this gap (Pérez-Gómez and Pelletier, 1997).

Designing in Virtual Environments (VE) may minimise this divergence. As a result of this designers are empowered to express, explore and convey their imagination with fewer differences (Dobson, 1998). Most researches on VE within the architectural context have focused on their use as presentation or simulation environments. Only some research is done that studies the use of VE for designing (Gero, 1999a).

Design is an activity that is greatly complex and influenced by numerous factors. It has been suggested that for this reason, the very different nature of VE may allow architects to create designs that differ from not VE aided design (Davidson and Campbell, 1996). This chapter examines the implications on architectural design by using immersive computer generated VE as the medium for design in a collaborative setting.

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2 Related Research

Architectural design within VE has been widely used as a method of design simulation and presentation. Educational and professional settings employ VE successfully to study, communicate and present architectural design. The rapid development of digital tools during the past decades had profound impacts on the architectural education and the way how architects create, converse or appreciate 3D spatial environments (Koutamanis, 1999). Numerous publications illustrate the impact digital media had on architectural design and propose solutions for multi-media design studios and how to make use of VE (Maver, 2002). Dave (1995) investigated distributed design studio. Wenz and Hirschberg (1997) studied collaborative design within remote collaboration, while Hirschberg et al. (1999) analyzed pattern of communication within digital design studios. VE often became a presentation tool only to assess design alternatives and final design solutions (Achten et al., 1999). Yet, they did not look into the comprehension and conception of design within Immersive VE (IVE).

One particular form of design studio emerged in the early nineties that investigated various possibilities digital media and VE can offer to the learning and the exploring of architectural design. These so called *Virtual Design Studios* (VDS) defined virtuality as acting while physically distant or as acting by employing digital tools (Maher et al., 2000). Yet virtual did not refer to an IVE. Instead, VE were established by the choice of design (Achten, 2001), way of communication (Schmitt, 1997) or tools (Kurmann, 1995; Regenbrecht et al., 2000; de Vries and Achten, 2002). Yet a significant potential of research remained unexplored. Within the context of a VDS, a real immersion into a VE could be used for designing.

3 Architectural Design in Virtual Environments

The question of describing architectural design itself is central to any debate about the use of VE to support architectural design. There are many definitions of design. Lawson (1980, 1994) for example defines architectural design as an act of creation, while Dorst (1996) tries to address more general definition of design related to any action within an artistic intention. Others have considered the difference between design in physical worlds and design in VE (Bridges and Charitos, 1997).

Obviously, there are significant differences between manual drawings, sketches, paintings, etc. and images and models produced using CAD systems. Since the introduction of CAAD by Sutherland's Sketchpad (Sutherland, 1963), extensive research has been undertaken to explore various possibilities and potentials of different media and realms (Goldschmidt, 1991; Schön and Wiggins, 1992; Robbins, 1994; Goel, 1995; Lawson and Loke, 1997; Suwa and Tversky, 1997; Verstijnen et al., 1998; Schnabel et al., 2007).

Nevertheless in all descriptions of the design development, we find in common the activities of cogitation, expression/modelling and communication/testing (Broadbent, 1973). Without engaging in discussion about computer-mediated cogitation or speculating on what the results of VE supported design may be, it is obvious that the tool used is affecting the process and outcome of a design. The impact of the medium of representation on the content has been the subject of many studies. Marshall McLuhan (1964) proclaimed that 'the medium is the message'. In the same way 2D drawings had a significant impact on architectural design in the 15th century, hence it is logical to assume that digital media, especially IVE, have a considerable impact on architects' ability to conceive, understand and communicate spatial environments.

3.1 Initial Design Stages

VE play an increasing role in architectural design especially during early design activities (Bertol and Foell, 1997). Equipment and software to engage in VE are easily available and particularly affordable. However, not sufficient attention has been paid to the results and possibilities of architectural design in IVE (Stuart, 1996). Lessons learnt from academic contexts have been employed in various commercial settings within the creative industries. A few corporations and architectural companies make use of global locations, sharing of resources and work force. This collaboration goes beyond video-conferencing or file-sharing and includes shared design sessions and expertise consultation (Burry et al., 2001).

The overall dimensions of an architect's final 'product' as well as constraints of resources make it usually inevitable that architects communicate and express their intentions with the help of models. Architects can make use of real and virtual instruments to translate and communicate their designs in these mixed realms (Schnabel, 2009a). In recent years, computer generated VE are increasingly used as a device of communication and presentation of design intentions (Leach, 2002). VE are employed successfully to study, communicate and present architectural design (Bertol and Foell, 1997). However, according to Maze (2002), IVE are seldom used in initial design stages for creation, development, form finding and collaboration of architectural design.

3.2 Computer Supported Collaborative Design - Virtual Design Studio

As Kvan (1999) argues, Computer Supported Collaborative Design (CSCD) can enhance the exploration of ideas and their communication. Kalay (1998) describes the difficult situation of architectural ventures that employ CSCD methods. Integrated projects are undertaken by fragmented teams, leading to decreased performance of both processes and products. Virtual Design Studios (VDS, 1993-2002) have been widely used in the last decade as an environment for architectural design teaching. Immersion has not been used for design interaction, although shared immersive virtual spaces have been employed for design reviews (Davidson and Campbell, 1996). The next logical step to develop the VDS was, therefore, to establish joint design sessions where users can collaboratively create, interpret and communicate design ideas within an IVE and to examine if this context offers any new opportunities or solutions to problems encountered. Mitchell (1995) argued that there is indeed a need for an ongoing evolution of the

VDS towards a fully integrated studio where the borderlines between realms are dismantled.

4 Designing in Virtual Environments

The objective of our research was to identify how designers perceive architectural space within VE by looking at the creation, interpretation and communication of architectural design in a collaborative setting. In order to investigate the relative effectiveness of IVE, we conducted a series of experiments. With this, we want to understand if form comprehension and form finding is enhanced within VE activity (Schnabel, 2004).

Therefore, we set up an experiment, a design studio that enabled students to design within a VE that imbeds immersive tools into a broad context of CSCD. The experiment engaged students in typical architectural design contexts. We were then able to draw conclusions that are relevant to the praxis of the architectural design profession. The Studio Experiment, called ‘Virtual Environment Design Studio’ (VeDS), look into conditions of architectural design, collaboration, communication, understanding and re-representation within a setting of an IVE design studio. It simulates a normal design process within the architectural profession by its nature (Schnabel and Kvan, 2003).

We explored factors influencing designers during the design process and investigated the relationship of 3D space within VE versus the physical realm.

As already mentioned above, instruments and designers have an influence on the outcome. Hence we engaged students of both genders of the master’s programmes at the departments of architecture of the University of Hong Kong (HKU) and the Bauhaus University Weimar (BUW), Germany.

5 Immersive Virtual Environment Design Studio (VeDS)

The design studio is the established context for architectural learning. Collaborative learning and designing have been demonstrated to support effective learning in architectural design (Kvan et al., 1999). As a result, we explored in more detail how a VDS, that is truly virtual and designers are immersed into a VE, affect the process and outcome of design (Schnabel, 2002).

We sought tasks that engaged designers at different levels of complexity within VE. Thus, we decided upon the design of a commercial helicopter landing station in the urban setting of Hong Kong. This task required students to work in three dimensions at all times and to fully navigate a 3D (not 2.5D) space. The IVE interface did intentionally not to allow a detailed modelling and users could therefore only establish initial layout of solid and voids in order to generate their design proposals. In this assignment, the designer could work in a virtual model of Hong Kong from the viewpoint of the pilot flying to and from the site or of the passenger waiting at the helipad to embark.

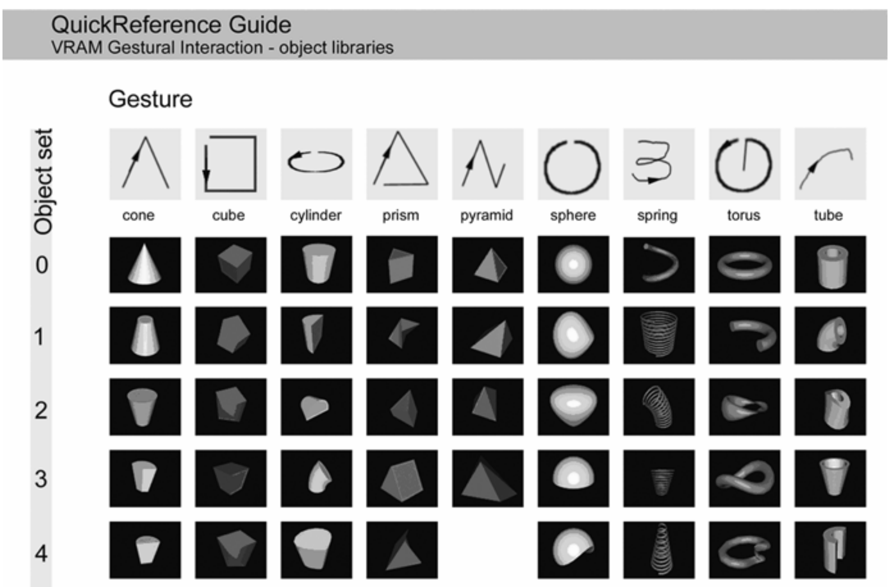
The assignment for teams was split in two parts, one for each team: either the land- or the airside of the helipad. Additionally each part of the task had one static and formal, as well as one dynamic and path focused. Both of which had to be addressed in the design proposal (Table 1). The teams gained authority of a design area and at the same time had to negotiate with the remote team, who worked on another area of the same site. Both teams had to collaborate in order to reach a solution while they also had enough freedom to explore their own design aspirations.

Table 1. Design task distribution: Landside/Airside and Static/Dynamic

Team	Authority	Programme	Component
A	*Landside*	*Check-in/Waiting enclosure for passengers*	→ *static*
		Driveway/Parking	→ *dynamic*
B	*Airside*	*Control tower for Air controllers and tourists*	→ *static*
		Apron/Flying	→ *dynamic*

We modified the *Virtual Reality Architectural Modeller* (VRAM) software by Regenbrecht et al. (2000), and added new input features based on gestures, called *VRAM/G (Seichter, 2001).* Comparable to the input for handheld computers or PDA devices, users gestured with the *Stylus* and their movements were tracked by the tracking device and, via one computer, translated by VRAM/G into basic 3D primitives. Table 2 shows the reference guide of gestures that the software recognises and translates into relevant primitives. A set of object libraries allows variations of a set of primitives.

Table 2. Gesture Reference Guide



A second PC was used as communication channel, using *ICQ*-software, an internet browser *(Internet Explorer)*, a web-based database (Figure 1) as well as other presentation software *(AutoDesk 3DStudio VIZ* and *Adobe Photoshop).*

Fig. 1. Screenshots of Database of VeDS; Left: Overview of output by one of the teams; Right: Presentation of work in one phase with text annotation {http://courses.arch.hku.hk/vds/veds01/db}

As in a moderated discussion session where the microphone is passed to speakers, the *Stylus* was virtually passed between the teams on each remote side and the resultant design sketches were produced within the IVE in the course of the alternating sessions. To support the design process more fully, text communication was also provided (Wong and Kvan, 1999). In order to capture the design intent, we used a modified 'think aloud' methodology by establishing a design team of two participants at each end: one team member wears the HMD designing actively with the *Stylus* and the second team member takes notes and chats with the remote team via chat-lines conveying design intent and action. The remote team has the same pairing of one team member wearing the HMD and the other communicating via the text-channels. By this way, we created a 'mental unit', in which two designers form one 'design unit' (Figure 2).

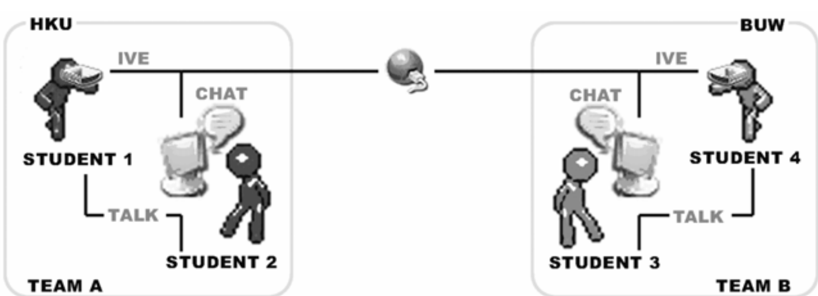


Fig. 2. 'Mental Unit': each side teams up in pairs to form one design unit

The designer using the HMD was free of all complex communication actions that could hinder the smooth flow of the act of design. While the other team member was only writing and corresponding with the remote team, a record of all communication was automatically generated. In this way, the text records provided a protocol to be analysed.

Although in the past our goal has been to engage in heterogeneous environments, with each participant using whatever equipment they wish, the problems of VE collaboration precluded such freedom. In this experiment, both universities employed the same configurations and used the same immersive VR equipment and all participants of these experiments received thorough instructions on the use of the equipment prior to the studio.

The design tasks were specified in order to present the students with assignments that are appropriate in scale, content and effort to the medium available. Special care was taken neither to favour nor to hinder the designers in creativity and translation of idea and result.

6 Discussion

This experiment showed that students using IVE engage actively in exploring space and volume in an inclusive manner. We validated that it is possible to successfully design, communicate and collaborate in IVE using an architectural design studio setting. Students communicated their spatial ideas using 3D objects that are accessible and interpretable by others in close relationship with the original design intent. This can either be very lively and similar to a sketch or as complex as a full design studio. Although it was possible that the teams would concede to the technical complexity of the system and the difficulty of working together, the teams did engage in collaborative work, building up, step by step, on the work of the efforts of team partners and preceding steps.

All teams have succeeded in designing a helipad and thus completed their tasks. The VeDS allowed a variety of design outcomes since every design studio has its own characteristic and every design-team their own approaches. Consequently, the results can be analysed in various ways. We explored the essence of the progress, form and communication of the students' works as described below. General speaking, the outcomes of the VeDS confirm that collaboration and designing within IVE lead to successful and valid architectural design schemes (Schnabel and Kvan, 2002).

6.1 Progress and Form

A review of the graphic results and digital models shows that students used the 3D design space actively. Volumes were created to represent design elements at all cases within the available 3D design space. Typically, a design created in a 2D space would have placed elements in plan with some raised in section/elevation to create 3D spaces. In the experiment, however, the students started 'drawing' the design elements at all points above the ground plane. Observation during the creation processes of the designs showed that participants did not use a 'bottom-to-top' (floor by floor), an 'inside-out' (function defines form) or 'outside-in' (form defines function) approach to their design. Mostly, students used an integrated design method by making use of all approaches almost at the same time. Being virtually inside the model, they sculpted their proposals, employing the flexibility of viewpoints offered in IVE. They explored the spatial impact of their design proposals in relation to existing forms and activities from outside and inside the model. They changed their viewpoint constantly from a general macro- or overview to a micro view of details in order to check spatial relationships and design proportion of the various elements of their design.

Although the input systems were crude and clumsy, users rapidly learnt to represent their design intent by using the available representational volumes: cubes, cylinders, cones and spheres. The objective of the assignment was to establish initial relationships of form, space, solid and void, in order to establish the overall gestalt of the design, while a detailed planning was not required.

The software offered a variety of library elements. Nevertheless, the teams predominantly made only use of the most basic set of library elements. Despite this constraint, these primitives offered students a significant range of design expression. This matches the simplicity of the software, its interface and its operation that did not require any complex menus or operational overheads or special setups. Similar to conventional design media, such as paper and pen, initial ideas were sketched freely into the space. Equivalent to a 2D medium, the various shapes symbolised both positive and negative representations of form and space or symbolised just reference points or other graphical elements. Viewers of the models, however, were able to understand this ambiguity of this 3D sketching. Yet the three-dimensionality of the primitives and the ability to change viewpoints and scale allowed the students to explore the space in a way that a sketch cannot offer.

In some cases, because of the lack of experience, problems with the hard- or software or the complexity of IVE, errors occurred, but often they were transformed into meaningful solutions, a design behaviour which is observed in other traditional 2D design environments as well (Schön and Wiggins, 1992). For example, instead of deleting unwanted elements, some teams chose to keep those elements and integrate them into their design. These 'errors' were then actively repeated to generate the desired outcome.

Each VDS merges the diverse backgrounds and skills of participants, their education, knowledge and understanding of architectural design. Subsequently these differences are expressed in the design. Since each team had authority over its own area, the VeDS allowed a distinct development of each team's design and if desired by the teams a clear distinction between the two design areas. In the course of the studio, it was necessary for students to agree on common readings as well as shared definitions of each team's working space, their borders and common elements and parts as well as working and design strategies.

Participants noted in the chat line communications that the resultant designs surprised them in their ingenuity and presentation. Constrained to their own preconceptions, they realised that working with the tool within IVE was not only easier than initially anticipated, but also have created outcomes that were superior to those their own skills enabled them within other design media. It appears that IVE allowed students to experience their ideas in ways that are different from non-immersive environments. They reported that the interaction of idea and creation was direct, that each stroke had an immediate impact on the design. They could not only understand easily the intentions of the other teams but also were inspired by the ease and freedom the tool and the environment offered them. For the students, it seemed that they communicated directly with their model and being part of it, instead of being an only distant designer. The students stated that this kind of design method has led to new forms and new arrangements. Nevertheless, for all students it is a novelty to design within IVE. Architectural design studios have not developed much further from conventional design methods despite the available technologies and innovative topics. Students are therefore not used to designing three-dimensionally and interactively, as it is possible with IVE.

6.2 Communication

The experiments proved that the teams communicated confidently as anticipated, by means of the set up, of which two participants formed one communication unit to correspond with the remote team. The teams intensively discussed issues of design, concepts and form. Due to the nature of the task and application, the groups had to formulate their intentions and discuss them with their remote partners in order to develop their scheme further. In addition, participants developed a personal interest in sharing their experience and creation with their colleagues and other teams.

In the analysis of the chat protocols showed only a few lines of navigation/orientation discussions. This suggests that participants could not only orientate themselves easily within VE, but they were also able to abstract and extract the design intent of the remote partner without much difficulty. Neither the tool nor the environment was an issue to talk about because both of them have blended into the design process harmoniously.

While the text records do not identify how or why the students were using the 3D space in these ways, we do find records of intense discussion about design, functions and concepts.

By referring to the images they saw in the model provided by their distant collaborators, students could engage in design discussion and development of the scheme. VE did change how the students developed and expressed their ideas. This new way was then communicated to the remote team using the design itself and discussion it on chat texts.

We noticed that participants from BUW tended to deal with more conceptual schemas while HKU students tended to be more factual, specific and they described ideas in tangible terms, possibly reflecting the educational characteristics of the two institutions. With such distinctions, it is notable that the VR environment supported these differences and the collaboration was successful.

7 Conclusion

We developed our experiments based on reported results of prior research in design collaboration and communication using real and virtual environments

(Schmitt, 1998). We carried out an architectural virtual design studio that took the issues of IVE into a complete architectural design scenario. Then we transferred our experiences of the VeDS to some experiments with abstract problem solving.

In a similar context Dave (1995) also confirms that VE is a constructive tool to support the design and communication process at least in establishing co-presence for a shared experience in spatial review. Yet how is this support extended to a design setting? Chat-protocols show participants remarking to each other that the collaborative experience was satisfying. That means, in IVE the exploration of space, volume and location is enhanced and site-specific problems are not only better recognised, but also possibilities are investigated. This is an improvement over other forms of design sharing that is analogous to the conclusion drawn by Campbell and Wells (1994).

Using a 2D medium to translate spatial ideas apparently reduces the exploration and communication of volume and space. Coherent to our findings, Dorta (2001) concludes that VE have significant impact on the activities of communicating 3D information within a design process caused by the impact of VE on the cognitive aspects of the design activity: the formation of 3D mental images, visual perception and mental work load. The results of the experiment support those findings. VE permit an enhanced understanding of spatial compositions. In other words, using VE as medium to design spatial 3D compositions, designers can pursue ideas with a smaller cognitive workload.

While users of VE can change their viewpoints and escape gravity, they also maintain the feeling of presence within the digital 3D models are generated with the intention of conveying overall design intentions similar to physical models, constructed to improve the perception of designs developed by drawings. As a result, IVE provide an immediate feedback to users that are not possible within CAD or traditional design media (Chen, 1995). It appears, therefore, that designers can work more three-dimensionally within VE since every object is experienced through movement and interaction. The design is created as a whole entity within space and not as a 2D planar representation. This possibility offers a different 'conversation' with the design that otherwise is not obvious or possible. In addition, spatial issues can be addressed in a manner akin to the real world. The design process becomes more immediate, in some aspects, with the tools available enhancing the translation of the designers' and users' mental intention into spatial objects and 3D design decisions. Subsequently these possibilities have an impact on the quality of the resulting design. The experiences seem valuable even in spite of the amount of technological overhead used and the abstract realism of VE.

Only in very recent years architectural design is evolving beyond the traditional language (Gruber et al., 2003). Architects discover new ways and different tools to communicate their design (Schnabel et al., 2007). Hereby VE can help them to explore and express ideas unlike traditional methods.

A similar phenomenon happens within the academic and educational environment. Less than a decade ago many schools of architecture did not allow students to deliver CAD drawings for design projects assuming that those limit the exploration and understanding of design. In fact, the early experiments in using the computer in the design process quite often failed only because of the restrictions of the available hard- and software. Today, students are familiar with CAD software even before they enter the university (Dokonal and Hirschberg, 2003). Still many questions remain unanswered and new questions arise in the relationship between architectural design and digital media. Architectural design is both an imagination and the ability to convey this idea. The teaching of architectural design has now the possibilities to make use of the advantages that VE can offer without loosing the qualities of the established conventional methods. Yet too often however, in using digital media and VE tools the students are more conversant than the teachers are. All those changed the dynamics of architectural education. Yet, this has to be reflected in how we teach.

Finally we recognised that the translation of design from VE into other media is potentially problematic (Gero, 1999b), suggesting that developments may be needed to facilitate the making of physical models. Similar to Gibson and Kvan's (2002) findings this suggests that other technologies such as rapid prototyping or automated construction methods may have a significant contribution to make to a design process that engages VE. Synergies between the different realms, media and technologies can be developed in a collaborative environment that fosters the evolution of new kinds of forms and structures.

This introduces a new way of designing and therefore fits well within existing paradigms (Mitchell, 1994). Wiener (1954) predicts the future merging of location and culture. Referring how physical and virtual architecture is an expression of cultural understanding. Both have their own properties, but both deal with the same matter. This will result in new understanding of architectural design and vice versa, this understanding influences the definition of architectural design (Schnabel, 2009b).

The potentials of VE are obvious and omni-present, yet they are not explored fully to their own capacities. As Maver (1973) postulates: "Design follows its own paradigms". Therefore, it evolves and re-establishes itself by its own developed expression.

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